

E-Commerce Sales Performance Analysis & Revenue Forecasting

End-to-end analytics project using Python, SQL, Databricks, Power BI and Machine Learning
Jan–Jun 2026 | 1,000 Orders | ₹27.62M Revenue

1. Business Problem & Objective

Analyze e-commerce sales performance across products, categories and locations.

Identify revenue trends, top-performing products and underperforming areas.

Understand changes in monthly revenue and sales contribution.

Build a short-term daily revenue forecasting model for planning.

Convert analytical findings into actionable business recommendations.

2. Dataset & Key Business Metrics

Simulated transaction-level e-commerce dataset covering January–June 2026

1,000

Orders

3,035

Units Sold

₹27.62M

Revenue

₹27,621

Average Order Value

Core fields: Order ID, Order Date, Product, Category, City, Region, Quantity, Price and Total Price.

Analysis performed at transaction, product, category, geographic and daily time-series levels.

3. End-to-End Analytics Workflow

Python → data quality checks, cleaning and exploratory analysis

SQL → KPIs, product/category analysis, geographic analysis and window functions

Databricks → curated data and Gold analytical tables using Delta

Feature Engineering → lag and rolling revenue features without target leakage

Machine Learning → baseline, Linear Regression and Random Forest comparison

Power BI → executive and regional sales dashboards

Business Insights → recommendations and short-term revenue expectations

4. Key Business Insights

Electronics generated approximately 78.2% of total revenue, making it the dominant category.

The top five products contributed about 88% of revenue, indicating strong revenue concentration.

Laptop was the highest-revenue product, followed by other major Electronics products.

Pune, Mumbai, Jaipur and Bangalore were among the leading cities, each contributing roughly ₹3.1M–₹3.3M.

Revenue peaked in January at approximately ₹5.21M.

Revenue declined by about 22% from May to June, with June revenue around ₹3.81M.

Books and Fashion had relatively low revenue contribution compared with Electronics.

5. Databricks Lakehouse & Analytical Layer

Created a curated sales layer from cleaned transaction data.

Built Gold tables for monthly, product, category, city and time-series analysis.

Created gold_daily_sales with daily revenue, orders, units, AOV and average selling price.

Created gold_sales_features for machine-learning-ready time-series features.

Used Delta tables to provide reusable analytical datasets for reporting and forecasting.

180 consecutive daily observations → 150 ML-ready observations after historical-feature preparation

6. Revenue Forecasting — Machine Learning

Target: daily total revenue.

Calendar features: day, day of week, week, month, quarter and weekend indicator.

Historical features: 1-, 7-, 14- and 30-day revenue lags.

Rolling features: 7-, 14- and 30-day historical revenue averages.

Used a chronological train-test split to respect the time-series nature of the problem.

Compared a previous-day baseline, Linear Regression and Random Forest.

7. Model Performance

Model	MAE	RMSE	MAPE
Previous-Day Baseline	₹121,610	₹161,198	262.83%
Linear Regression	₹99,847	₹116,240	195.86%
Random Forest	₹92,069	₹108,779	188.17%

Random Forest achieved the lowest MAE, RMSE and MAPE on the chronological test set.

Compared with the previous-day baseline, MAE improved by approximately 24.3% and RMSE by 32.5%.

MAPE is treated as supplementary because low-revenue days can create large percentage errors.

8. 7-Day Revenue Forecast

Date	Predicted Revenue
30 Jun 2026	₹312,523.78
1 Jul 2026	₹266,568.16
2 Jul 2026	₹242,149.56
3 Jul 2026	₹142,850.30
4 Jul 2026	₹106,184.20
5 Jul 2026	₹97,034.30
6 Jul 2026	₹100,033.94

₹1.27M
7-Day Expected Revenue

₹181K
Average Daily Forecast

₹312.5K
Highest Daily Forecast

9. Business Recommendations

Investigate the June revenue decline to identify the main drivers of the slowdown.

Monitor inventory and sales performance of high-revenue Electronics products.

Reduce dependency on a small number of top products by promoting mid-tier products.

Investigate underperforming cities for pricing, assortment, marketing or distribution opportunities.

Review low-contribution categories such as Books and Fashion for targeted growth opportunities.

Use orders, units, AOV and product contribution together rather than relying on unit volume alone.

Use the 7-day forecast as a planning input, not as a guaranteed revenue target.

10. Project Outcome

Built a complete analytics workflow from raw transactions to business recommendations.

Combined Python, SQL, Databricks, Power BI and machine learning in one project.

Identified major revenue concentration across categories, products and cities.

Built leakage-safe historical features and evaluated multiple forecasting approaches.

Selected Random Forest based on chronological test-set performance.

Generated a 7-day recursive revenue forecast to support short-term planning.

Demonstrated how analytics and forecasting can turn transaction data into actionable business insights.